

## ✅ ADVANCED NIX FLAKE CI/CD PIPELINE – Production Grade (v2.1.0)

Sovereign Machina Monorepo

[github.com/nodestadt/50V3R31GN-M4CH1N4](https://github.com/nodestadt/50V3R31GN-M4CH1N4)

This is the **complete, advanced** CI/CD setup aligned with your exact monorepo + patched Hermes fork workflow.

### Key Features of This Pipeline

- Submodule pinning validation on every run
- Full Nix flake check + build for all nodes
- Cachix binary cache for speed
- Matrix builds for Nodes A/B/C/D
- Deployment to live nodes via SSH + `nixos-rebuild`
- Security scanning (Trivy)
- Daily submodule audit with upstream change detection
- Manual dispatch for on-demand full rebuilds
- Strict failure on unpinned or dirty submodules

### 1. `.github/workflows/ci-pr.yml` (Runs on every PR and push to non-master branches)

YAML

```
name: CI - Pull Request & Branch

on:
  pull_request:
    branches: [ master ]
  push:
    branches-ignore: [ master ]

jobs:
  validate:
    runs-on: ubuntu-24.04
```

steps:

- uses: actions/checkout@v4  
with:
  - submodules: recursive
  - fetch-depth: 0
- name: Setup Nix  
uses: cachix/install-nix-action@v26  
with:
  - nix\_path: nixpkgs=channel:nixos-unstable
- name: Enable Cachix Cache  
uses: cachix/cachix-action@v15  
with:
  - name: nodestadt
  - authToken: '\${{ secrets.CACHIX\_AUTH\_TOKEN }}
- name: Verify Submodule Pins  
run: |
  - if git submodule status | grep -E '^-[U]'; then
  - echo "❌ Submodule not pinned or has conflicts"
  - exit 1
  - fi
  - echo "✅ All submodules properly pinned"
- name: Nix Flake Check  
run: nix flake check --all-systems
- name: Rust Linting & Testing  
run: |
  - nix develop -c cargo fmt --all -- --check
  - nix develop -c cargo clippy --all-targets --all-features -- -Dwarnings
  - nix develop -c cargo test --all --all-features
- name: Pretext HUD Build  
run: nix build .#pretext-hud
- name: Trivy Security Scan  
uses: aquasecurity/trivy-action@v0.18  
with:
  - scan-type: 'fs'
  - scan-ref: '.'

```
format: 'sarif'
output: 'trivy-results.sarif'
```

## 2. .github/workflows/ci-master.yml (Runs on merge to master + manual dispatch)

YAML

```
name: CI + Deploy - Master

on:
  push:
    branches: [ master ]
  workflow_dispatch:

jobs:
  build:
    runs-on: ubuntu-24.04
    strategy:
      matrix:
        node: [node-a, node-b, node-c, node-d]
    steps:
      - uses: actions/checkout@v4
        with:
          submodules: recursive
          fetch-depth: 0

      - name: Setup Nix + Cachix
        uses: cachix/install-nix-action@v26
        with:
          nix_path: nixpkgs=channel:nixos-unstable

      - name: Cachix Cache
        uses: cachix/cachix-action@v15
        with:
          name: nodestadt
          authToken: '${{ secrets.CACHIX_AUTH_TOKEN }}'

      - name: Build Node ${{ matrix.node }}
        run: nix build .#${{ matrix.node }} --print-build-logs
```

```

deploy:
  needs: build
  runs-on: ubuntu-24.04
  steps:
    - uses: actions/checkout@v4
      with:
        submodules: recursive

    - name: Setup Nix
      uses: cachix/install-nix-action@v26

    - name: Deploy to Nodes (Sequential)
      run: |
        for node in node-d node-c node-b node-a; do
          echo "Deploying $node..."
          nixos-rebuild switch --flake .#$node \
            --target-host root@$node.tailscale \
            --build-host root@$node.tailscale \
            --option builders '' || exit 1
        done

```

### 3. .github/workflows/submodule-audit.yml (Daily Audit)

YAML

```

name: Daily Submodule Audit & Security

on:
  schedule:
    - cron: '0 3 * * *'    # 3 AM UTC daily
  workflow_dispatch:

jobs:
  audit:
    runs-on: ubuntu-24.04
    steps:
      - uses: actions/checkout@v4
        with:
          submodules: recursive
          fetch-depth: 0

```

```

- name: Setup Nix
  uses: cachix/install-nix-action@v26

- name: Verify Submodule Integrity
  run: |
    git submodule status | grep -E '^-|U' && echo "❌ Submodule
issue detected" && exit 1 || echo "✅ Submodules clean"

- name: Check for Upstream Updates
  run: ./scripts/check-upstream-updates.sh

- name: Trivy Full Scan
  uses: aquasecurity/trivy-action@v0.18
  with:
    scan-type: 'fs'
    scan-ref: '.'
    format: 'table'

- name: Post Summary Comment
  if: always()
  uses: peter-evans/create-or-update-comment@v4
  with:
    issue-number: ${GITHUB_EVENT_PULL_REQUEST_NUMBER}
    body: |
      **Daily Submodule & Security Audit Complete**
      - Submodules: ${STEPS_VERIFY_OUTCOME}
      - Upstream updates detected: ${STEPS_UPSTREAM_OUTCOME}

```

#### 4. SUPPORTING SCRIPT – scripts/check-upstream-updates.sh

Place this in the monorepo root:

```
Bash
```

```
#!/usr/bin/env bash
echo "Checking for upstream updates in submodules..."

for submodule in $(git submodule status | awk '{print $2}'); do
    cd "$submodule" || continue
    git fetch origin main --quiet 2>/dev/null || true
    if [ "$(git rev-parse HEAD)" != "$(git rev-parse origin/main
2>/dev/null)" ]; then
        echo "⚠️ $submodule has upstream updates available"
    fi
    cd - > /dev/null
done

echo "Submodule audit complete."
```

Make it executable: `chmod +x scripts/check-upstream-updates.sh`

## ARCHITECT SIGN-OFF

This is a complete, advanced Nix Flake CI/CD pipeline that respects your monorepo + patched Hermes fork workflow.

- Strict submodule pinning enforcement
- Full Nix flake builds for all nodes
- Secure deployment via SSH
- Daily audit for upstream drift
- No automatic upstream syncs (deliberate only)

The pipeline is ready to be committed to `.github/workflows/`.

## Next Priority (Your Call)

Reply with the number:

1. **Add these CI/CD files** to the monorepo and trigger first run.
2. **Test plan** for the full CI/CD pipeline.
3. **Enhanced security rules** (signed commits, secret scanning, etc.).

4. **All three** as one package.

What would you like to initiate?